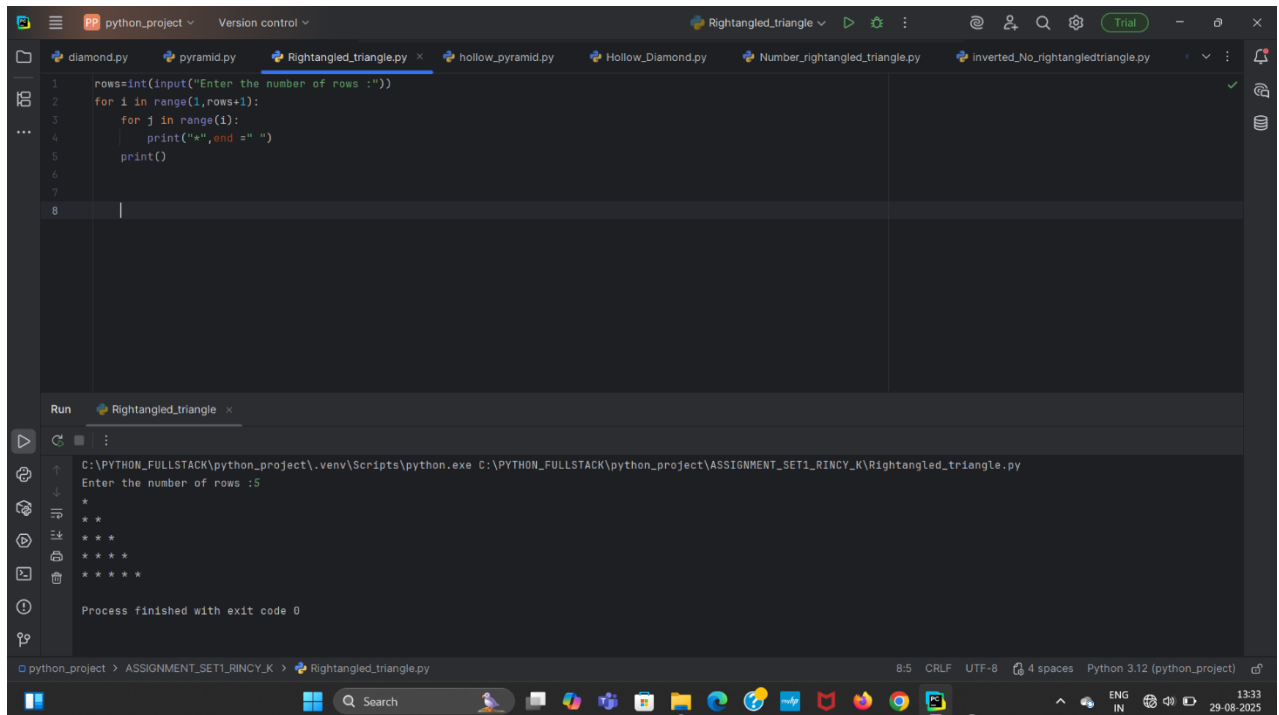


PYTHON PATTERN PROGRAMS

RINCY K

1.Right-Angled Triangle



The screenshot shows a Python IDE with a file explorer on the left containing files like diamond.py, pyramid.py, Rightangled_triangle.py, hollow_pyramid.py, Hollow_Diamond.py, Number_rightangled_triangle.py, and inverted_No_rightangledtriangle.py. The main editor displays the code for Rightangled_triangle.py:

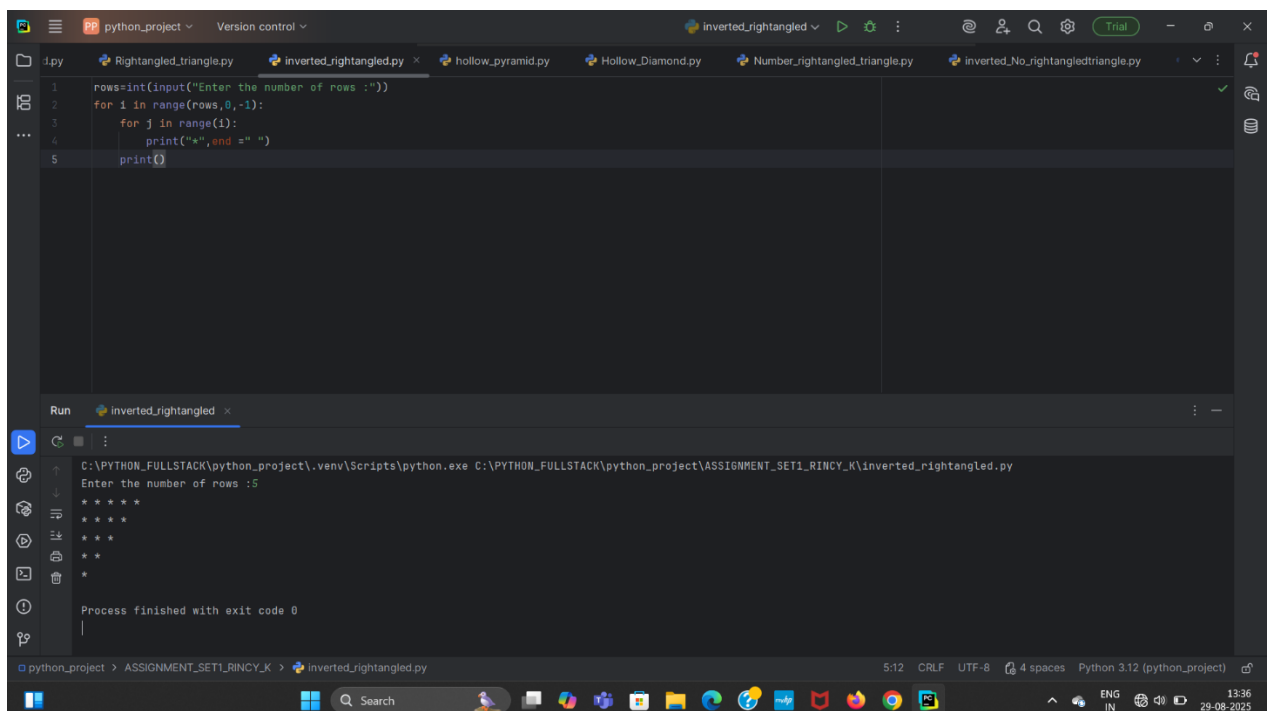
```
1 rows=int(input("Enter the number of rows :"))
2 for i in range(1,rows+1):
3     for j in range(1):
4         print("*",end=" ")
5     print()
6
7
8
```

The Run console shows the execution of Rightangled_triangle.py. The user entered 5 for the number of rows, and the output is a right-angled triangle of asterisks:

```
Enter the number of rows :5
*
*
*
*
*

Process finished with exit code 0
```

2.Inverted Right-Angled Triangle



The screenshot shows a Python IDE with a file explorer on the left containing files like Rightangled_triangle.py, inverted_rightangled.py, hollow_pyramid.py, Hollow_Diamond.py, Number_rightangled_triangle.py, and inverted_No_rightangledtriangle.py. The main editor displays the code for inverted_rightangled.py:

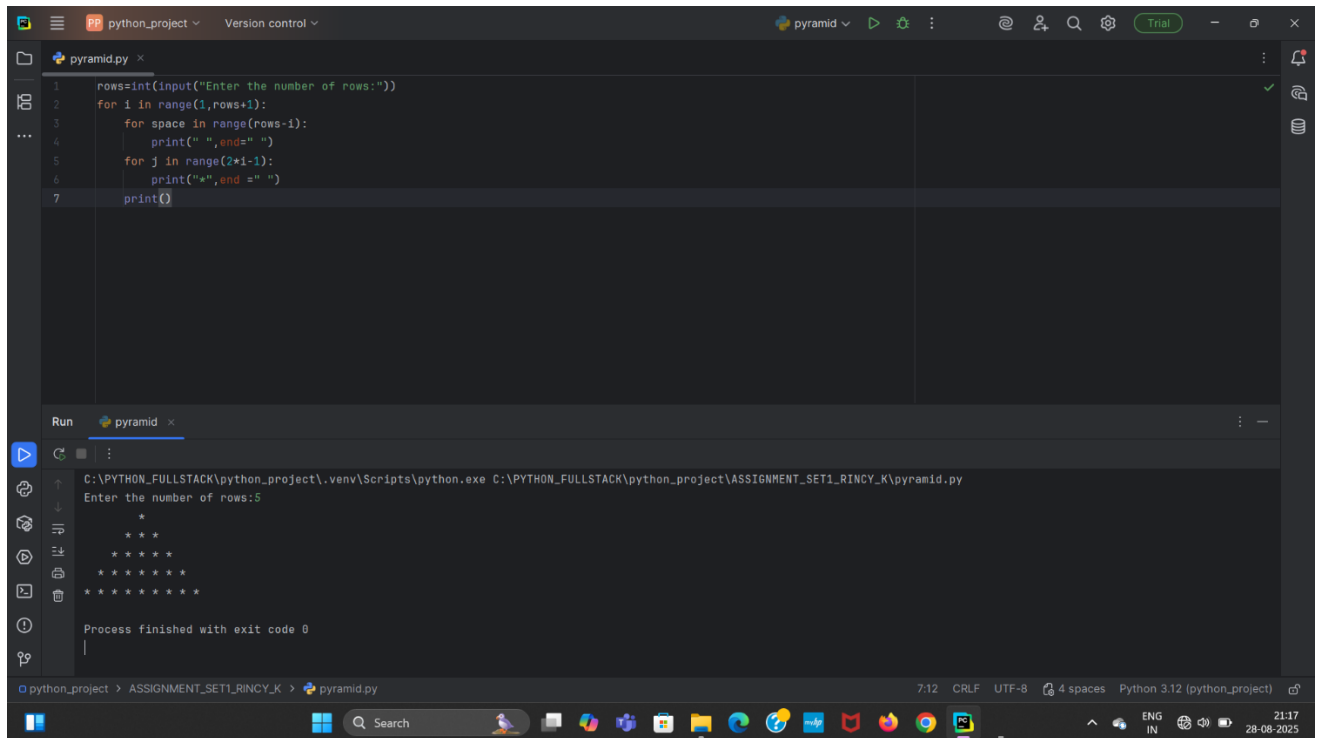
```
1 rows=int(input("Enter the number of rows :"))
2 for i in range(rows,0,-1):
3     for j in range(1):
4         print("*",end=" ")
5     print()
6
```

The Run console shows the execution of inverted_rightangled.py. The user entered 5 for the number of rows, and the output is an inverted right-angled triangle of asterisks:

```
Enter the number of rows :5
*****
*****
***
**
*

Process finished with exit code 0
```

3. Pyramid



The screenshot shows a code editor with a file named `pyramid.py`. The code is as follows:

```
1 rows=int(input("Enter the number of rows:"))
2 for i in range(1,rows+1):
3     for space in range(rows-i):
4         print(" ",end=" ")
5     for j in range(2*i-1):
6         print("*",end=" ")
7     print()
```

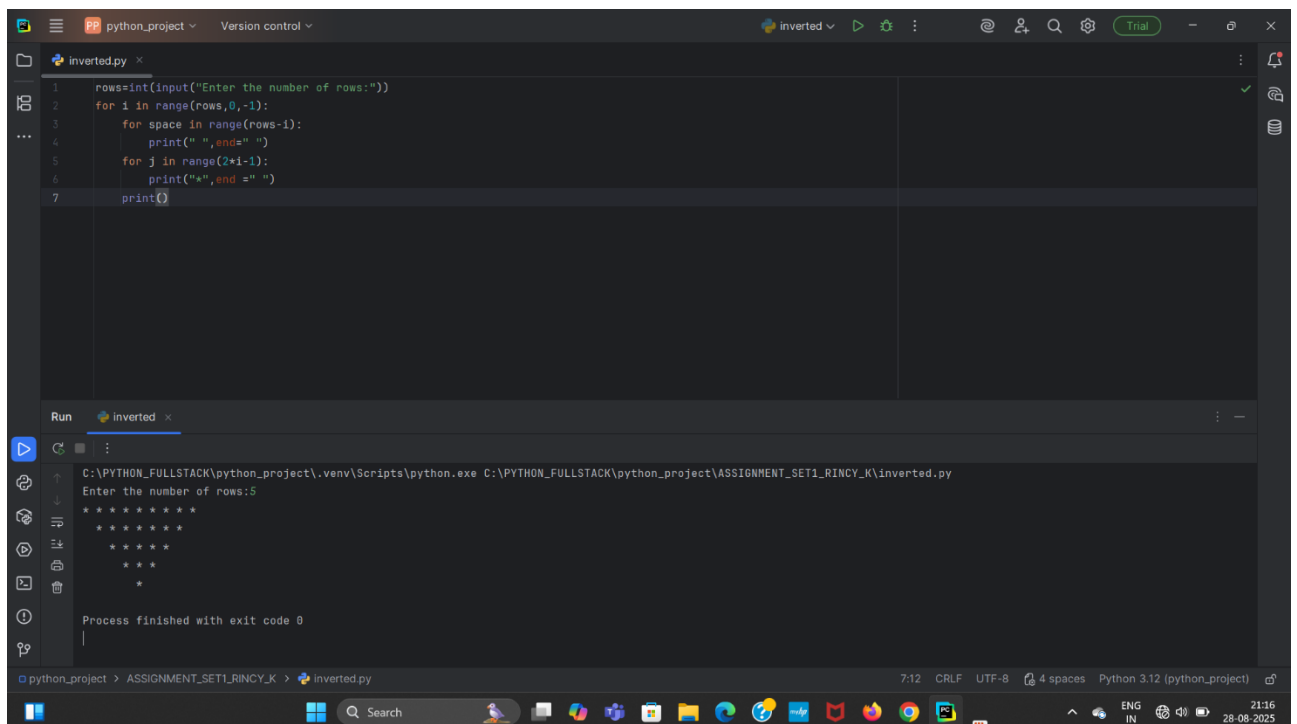
The Run window shows the output of the program:

```
C:\PYTHON_FULLSTACK\python_project\.venv\Scripts\python.exe C:\PYTHON_FULLSTACK\python_project\ASSIGNMENT_SET1_RINCY_K\pyramid.py
Enter the number of rows:5
      *
     **
    ***
   ****
  *****
 *****

Process finished with exit code 0
```

The status bar at the bottom indicates the file encoding is UTF-8, 4 spaces, and Python 3.12.

4. Inverted Pyramid



The screenshot shows a code editor with a file named `inverted.py`. The code is as follows:

```
1 rows=int(input("Enter the number of rows:"))
2 for i in range(rows,0,-1):
3     for space in range(rows-i):
4         print(" ",end=" ")
5     for j in range(2*i-1):
6         print("*",end=" ")
7     print()
```

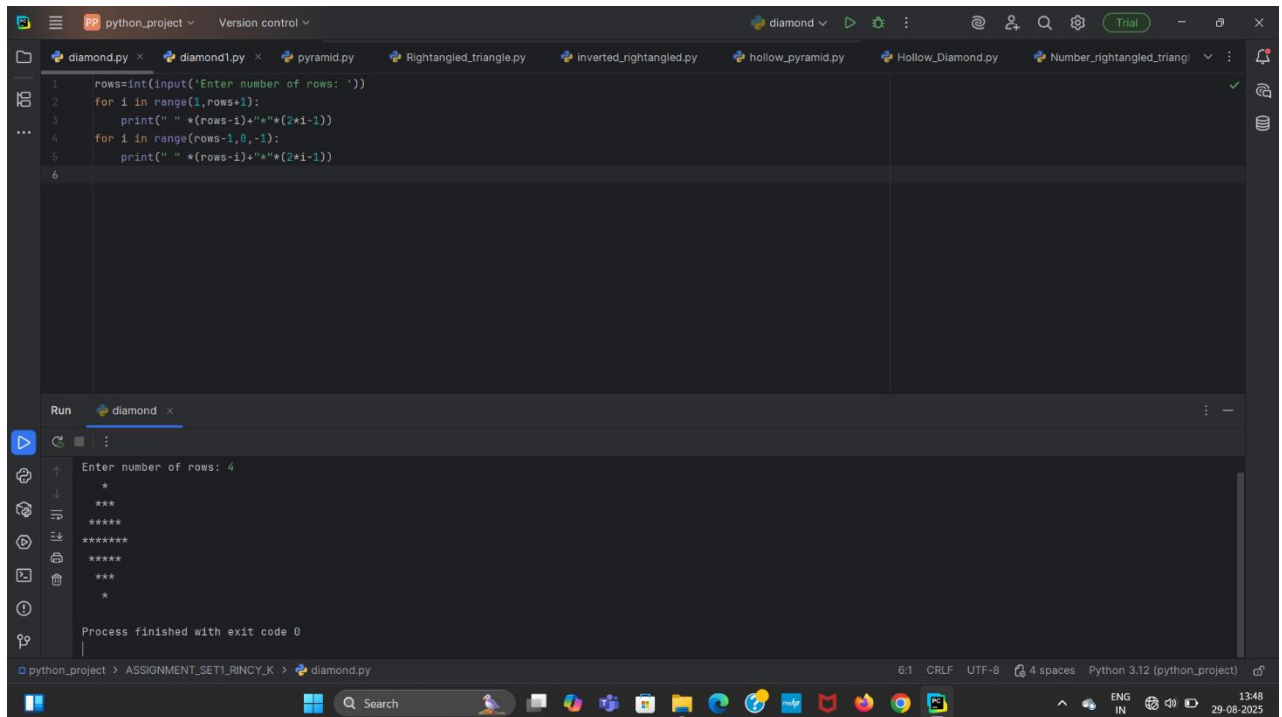
The Run window shows the output of the program:

```
C:\PYTHON_FULLSTACK\python_project\.venv\Scripts\python.exe C:\PYTHON_FULLSTACK\python_project\ASSIGNMENT_SET1_RINCY_K\inverted.py
Enter the number of rows:5
*****
 ****
  ***
   **
    *

Process finished with exit code 0
```

The status bar at the bottom indicates the file encoding is UTF-8, 4 spaces, and Python 3.12.

5.Diamond



The screenshot shows a Python IDE with a file named `diamond.py`. The code prompts the user to enter the number of rows, which is 4. The output displays a diamond pattern of asterisks. The pattern consists of 4 rows: the first row has 1 asterisk, the second has 3, the third has 5, and the fourth has 3. The spaces are used to center the asterisks.

```
rows=int(input("Enter number of rows: "))
for i in range(1,rows+1):
    print(" "*(rows-i)+"*"(2*i-1))
for i in range(rows-1,0,-1):
    print(" "*(rows-i)+"*"(2*i-1))
```

Run diamond x

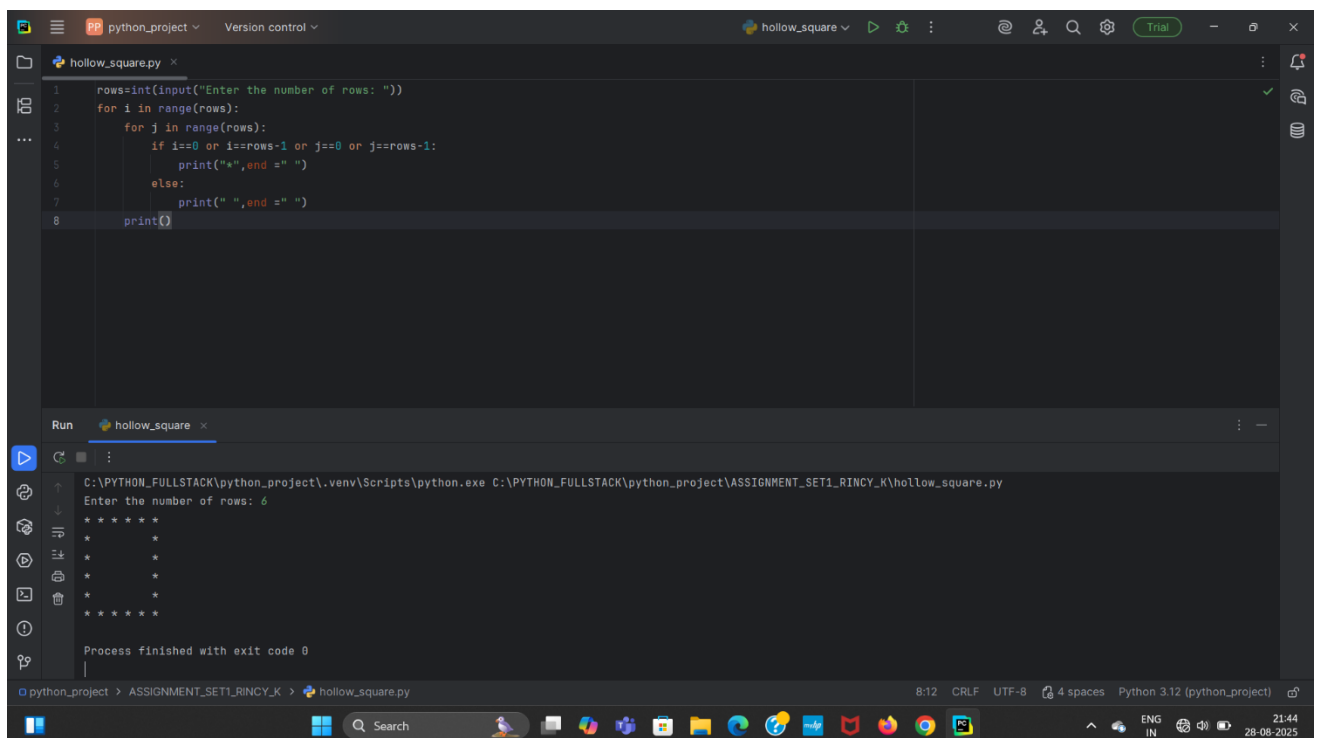
Enter number of rows: 4

```

 *
***
*****
***
 *
```

Process finished with exit code 0

6.Hollow Square



The screenshot shows a Python IDE with a file named `hollow_square.py`. The code prompts the user to enter the number of rows, which is 6. The output displays a hollow square pattern of asterisks. The pattern consists of 6 rows and 6 columns. The first and last rows are solid, and the first and last columns are solid, with the interior being empty.

```
rows=int(input("Enter the number of rows: "))
for i in range(rows):
    for j in range(rows):
        if i==0 or i==rows-1 or j==0 or j==rows-1:
            print("*",end=" ")
        else:
            print(" ",end=" ")
    print()
```

Run hollow_square x

C:\PYTHON_FULLSTACK\python_project\.venv\Scripts\python.exe C:\PYTHON_FULLSTACK\python_project\ASSIGNMENT_SET1_RINCY_K\hollow_square.py

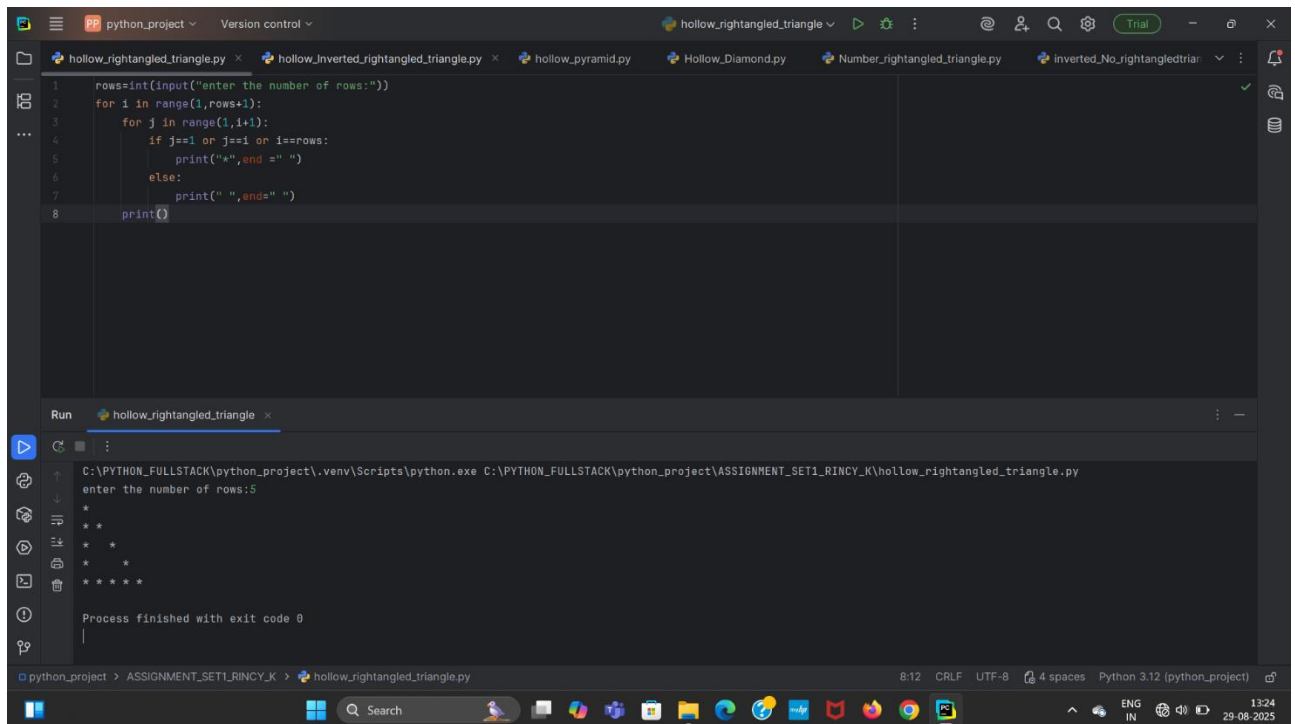
Enter the number of rows: 6

```

*****
*      *
*      *
*      *
*      *
*****
```

Process finished with exit code 0

7.Hollow Right-Angled Triangle



The screenshot shows a Python IDE with a file named `hollow_rightangled_triangle.py`. The code prompts the user to enter the number of rows, which is 5. The output shows a hollow right-angled triangle with 5 rows. The first row has 5 stars, and subsequent rows have stars at the beginning and end, with spaces in between.

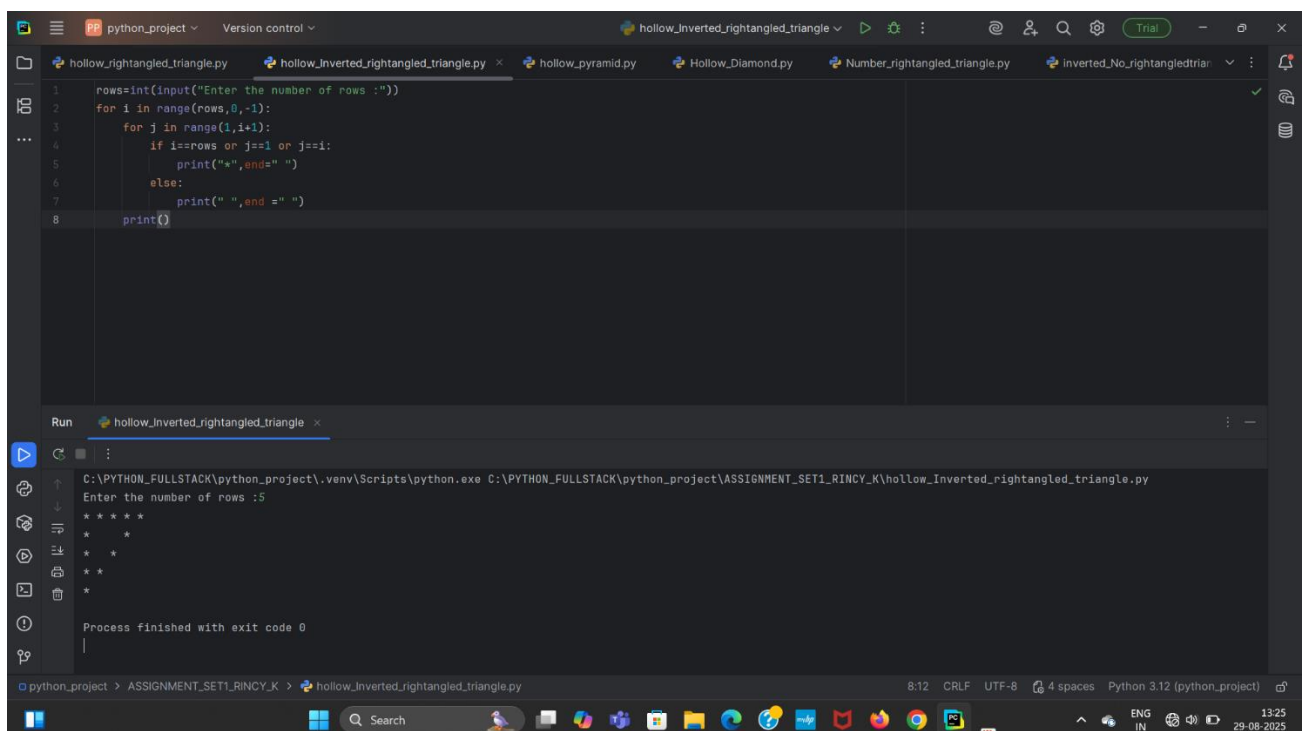
```
1 rows=int(input("enter the number of rows:"))
2 for i in range(1,rows+1):
3     for j in range(1,i+1):
4         if j==1 or j==i or i==rows:
5             print("*",end=" ")
6         else:
7             print(" ",end=" ")
8     print()
```

Run `hollow_rightangled_triangle`

```
C:\PYTHON_FULLSTACK\python_project\.venv\Scripts\python.exe C:\PYTHON_FULLSTACK\python_project\ASSIGNMENT_SET1_RINCY_K\hollow_rightangled_triangle.py
enter the number of rows:5
*
* *
* * *
* * * *
* * * * *
```

Process finished with exit code 0

8.Hollow Inverted Right-Angled Triangle



The screenshot shows a Python IDE with a file named `hollow_inverted_rightangled_triangle.py`. The code prompts the user to enter the number of rows, which is 5. The output shows a hollow inverted right-angled triangle with 5 rows. The first row has 5 stars, and subsequent rows have stars at the beginning and end, with spaces in between.

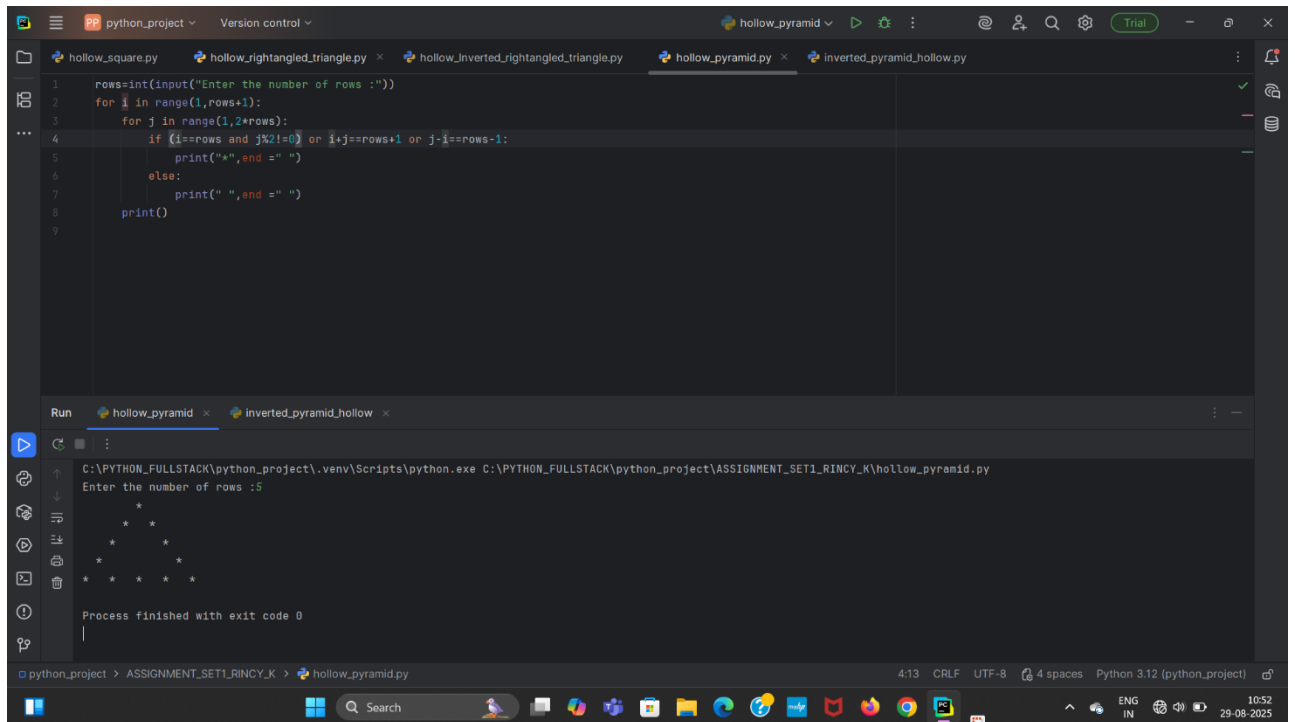
```
1 rows=int(input("Enter the number of rows :"))
2 for i in range(rows,0,-1):
3     for j in range(1,i+1):
4         if i==rows or j==1 or j==i:
5             print("*",end=" ")
6         else:
7             print(" ",end=" ")
8     print()
```

Run `hollow_inverted_rightangled_triangle`

```
C:\PYTHON_FULLSTACK\python_project\.venv\Scripts\python.exe C:\PYTHON_FULLSTACK\python_project\ASSIGNMENT_SET1_RINCY_K\hollow_inverted_rightangled_triangle.py
Enter the number of rows :5
* * * * *
* * *
* *
*
* * * * *
```

Process finished with exit code 0

9.Hollow Pyramid



The screenshot shows a Python IDE with a file explorer on the left containing files: `hollow_square.py`, `hollow_rightangled_triangle.py`, `hollow_inverted_rightangled_triangle.py`, `hollow_pyramid.py`, and `inverted_pyramid_hollow.py`. The `hollow_pyramid.py` file is open, displaying the following code:

```
1 rows=int(input("Enter the number of rows :"))
2 for i in range(1,rows+1):
3     for j in range(1,2*rows):
4         if (i==rows and j%2!=0) or i+j==rows+1 or j-i==rows-1:
5             print("*",end=" ")
6         else:
7             print(" ",end=" ")
8     print()
9
```

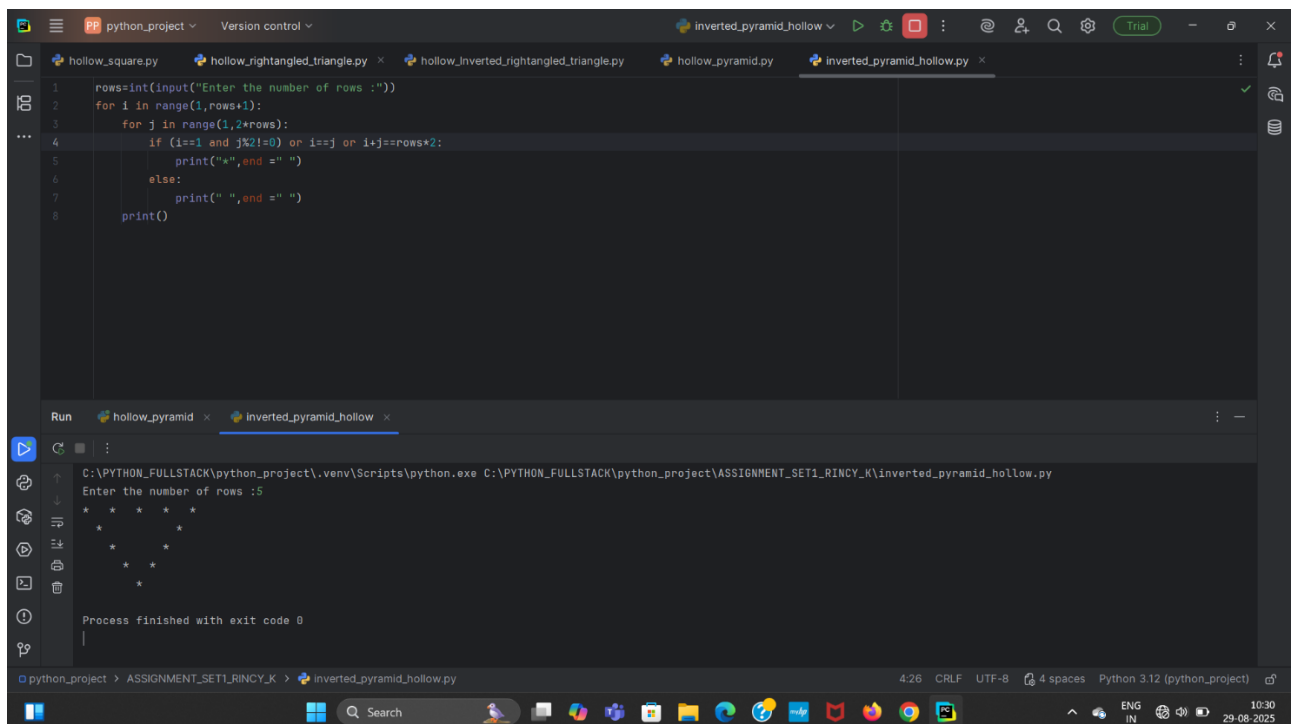
The Run console at the bottom shows the execution of `hollow_pyramid.py` with the input "5". The output is a hollow pyramid shape:

```

      *
     * *
    * * *
   * * * *
  * * * * *
```

The process finished with exit code 0. The status bar at the bottom indicates the file is `hollow_pyramid.py` with settings: 4:13, CRLF, UTF-8, 4 spaces, Python 3.12 (python_project).

10.Hollow Inverted Pyramid



The screenshot shows the same Python IDE with the `inverted_pyramid_hollow.py` file open. The code is as follows:

```
1 rows=int(input("Enter the number of rows :"))
2 for i in range(1,rows+1):
3     for j in range(1,2*rows):
4         if (i==1 and j%2!=0) or i==j or i+j==rows*2:
5             print("*",end=" ")
6         else:
7             print(" ",end=" ")
8     print()
9
```

The Run console shows the execution of `inverted_pyramid_hollow.py` with the input "5". The output is a hollow inverted pyramid shape:

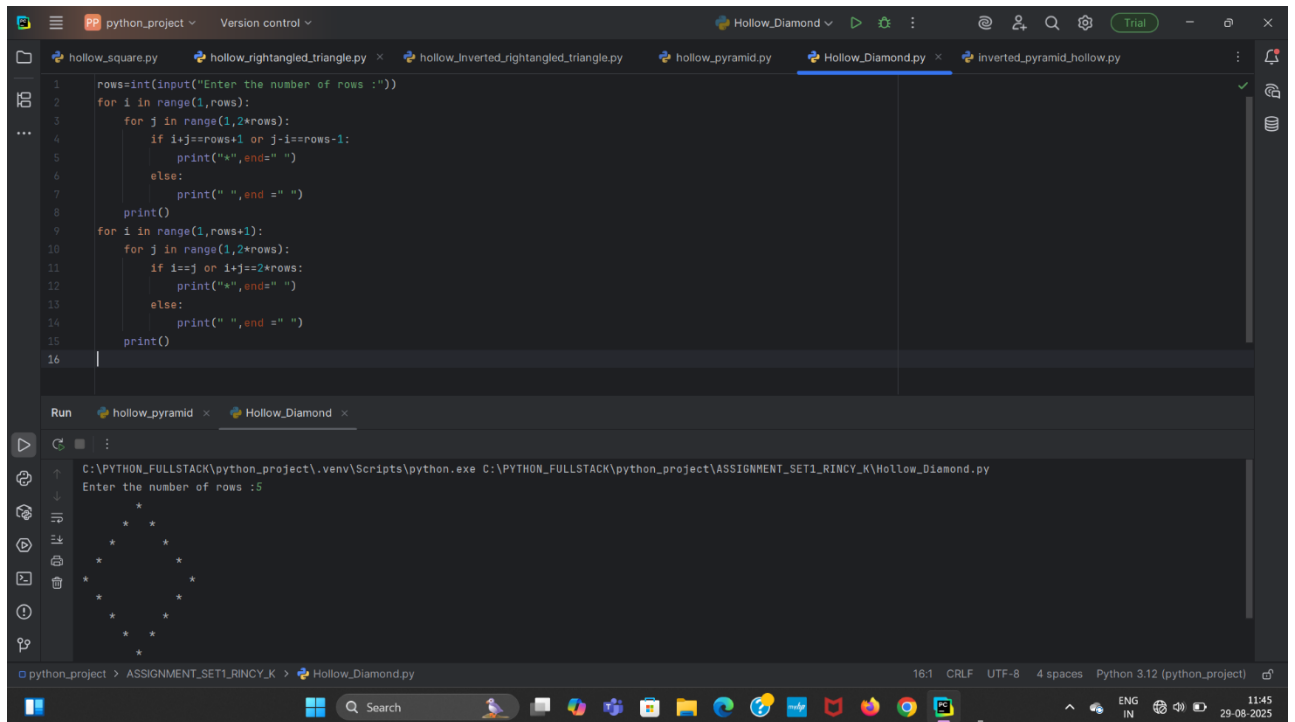
```

* * * * *
 * * * *
  * * *
   * *
    *

```

The process finished with exit code 0. The status bar at the bottom indicates the file is `inverted_pyramid_hollow.py` with settings: 4:26, CRLF, UTF-8, 4 spaces, Python 3.12 (python_project).

11.Hollow Diamond (Row = 5)



The screenshot shows a Python IDE with a file named `Hollow_Diamond.py`. The code prompts the user to enter the number of rows, which is 5. The output is a hollow diamond pattern of asterisks.

```
1 rows=int(input("Enter the number of rows :"))
2 for i in range(1,rows):
3     for j in range(1,2*rows):
4         if i+j==rows+1 or j-i==rows-1:
5             print("*",end=" ")
6         else:
7             print(" ",end=" ")
8     print()
9 for i in range(1,rows+1):
10    for j in range(1,2*rows):
11        if i==j or i+j==2*rows:
12            print("*",end=" ")
13        else:
14            print(" ",end=" ")
15    print()
16
```

Run `hollow_pyramid` `Hollow_Diamond`

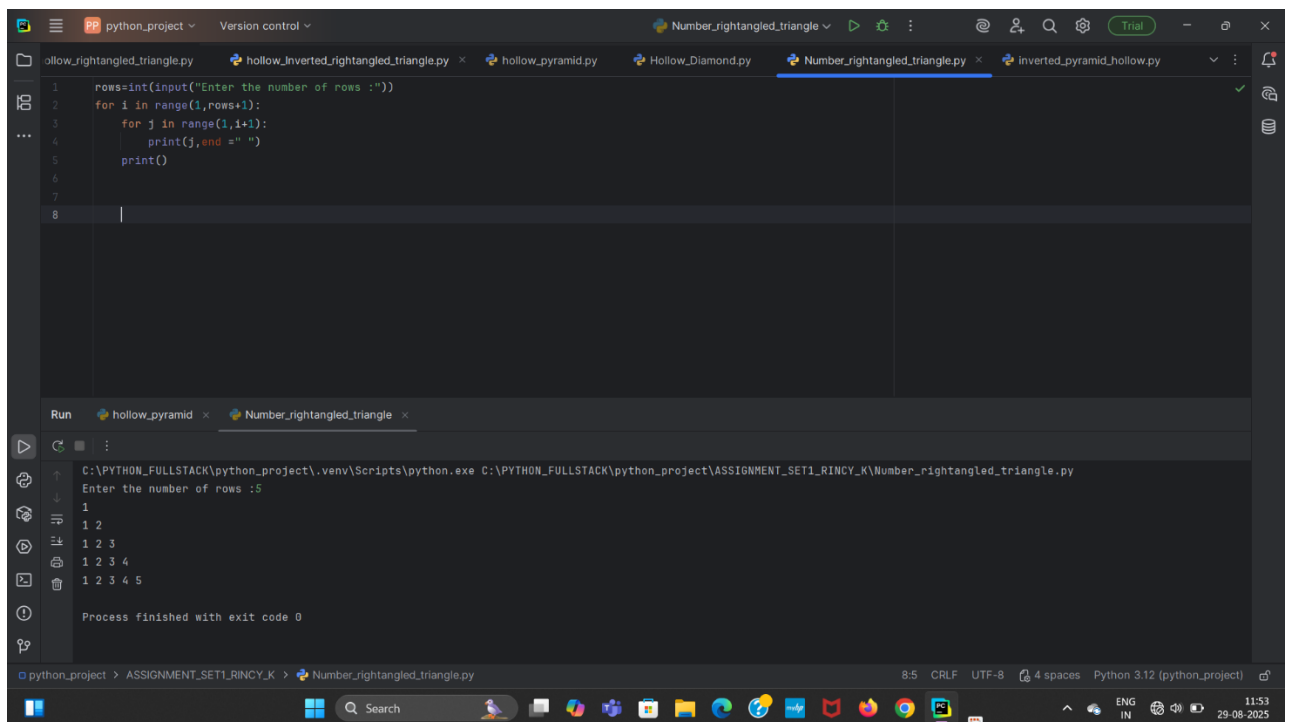
C:\PYTHON_FULLSTACK\python_project\.venv\Scripts\python.exe C:\PYTHON_FULLSTACK\python_project\ASSIGNMENT_SET1_RINCY_K\Hollow_Diamond.py

Enter the number of rows :5

```
  *
 * *
*   *
 *   *
  *   *
```

python_project > ASSIGNMENT_SET1_RINCY_K > Hollow_Diamond.py 16:1 CRLF UTF-8 4 spaces Python 3.12 (python_project) 11:45 29-08-2025

12.Number Right Angled Triangle



The screenshot shows a Python IDE with a file named `Number_rightangled_triangle.py`. The code prompts the user to enter the number of rows, which is 5. The output is a right-angled triangle of numbers.

```
1 rows=int(input("Enter the number of rows :"))
2 for i in range(1,rows+1):
3     for j in range(1,i+1):
4         print(j,end=" ")
5     print()
6
7
8
```

Run `hollow_pyramid` `Number_rightangled_triangle`

C:\PYTHON_FULLSTACK\python_project\.venv\Scripts\python.exe C:\PYTHON_FULLSTACK\python_project\ASSIGNMENT_SET1_RINCY_K\Number_rightangled_triangle.py

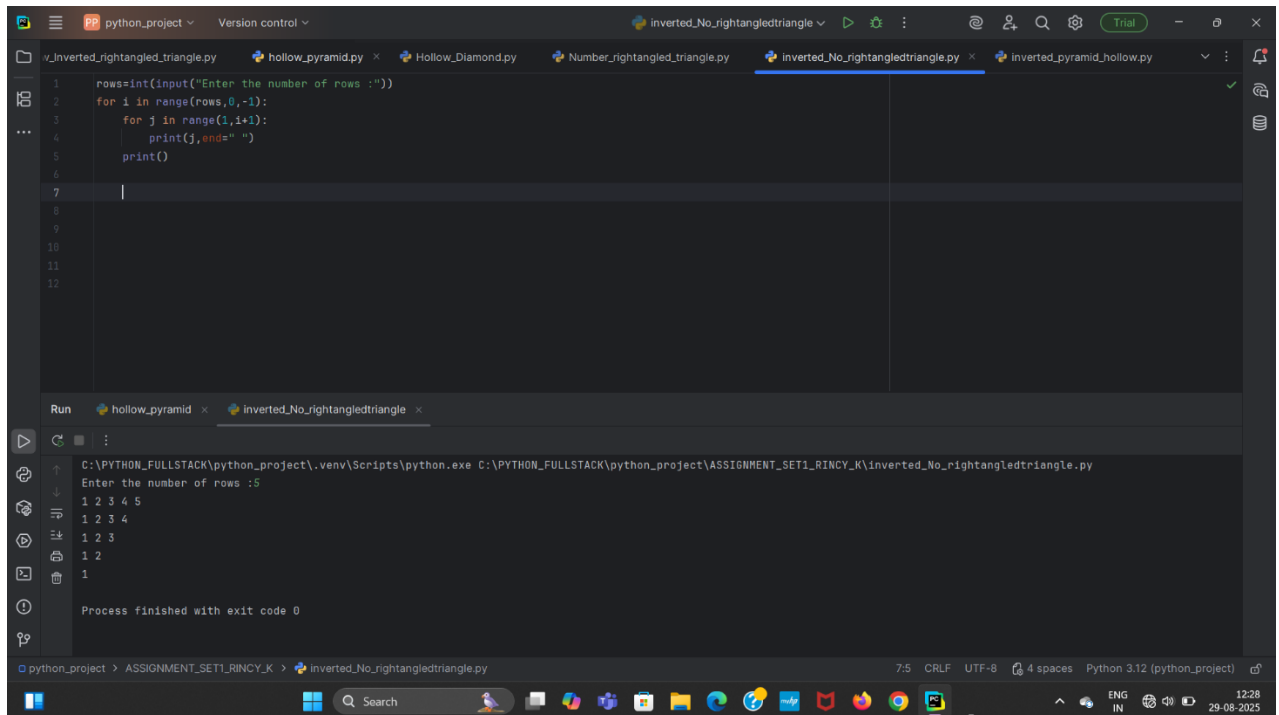
Enter the number of rows :5

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Process finished with exit code 0

python_project > ASSIGNMENT_SET1_RINCY_K > Number_rightangled_triangle.py 8:5 CRLF UTF-8 4 spaces Python 3.12 (python_project) 11:53 29-08-2025

13. Inverted Number Right Angled Triangle



The screenshot shows a Python IDE with a file named `inverted_No_rightangledtriangle.py`. The code prompts the user to enter the number of rows, and when 5 is entered, it prints an inverted number right-angled triangle.

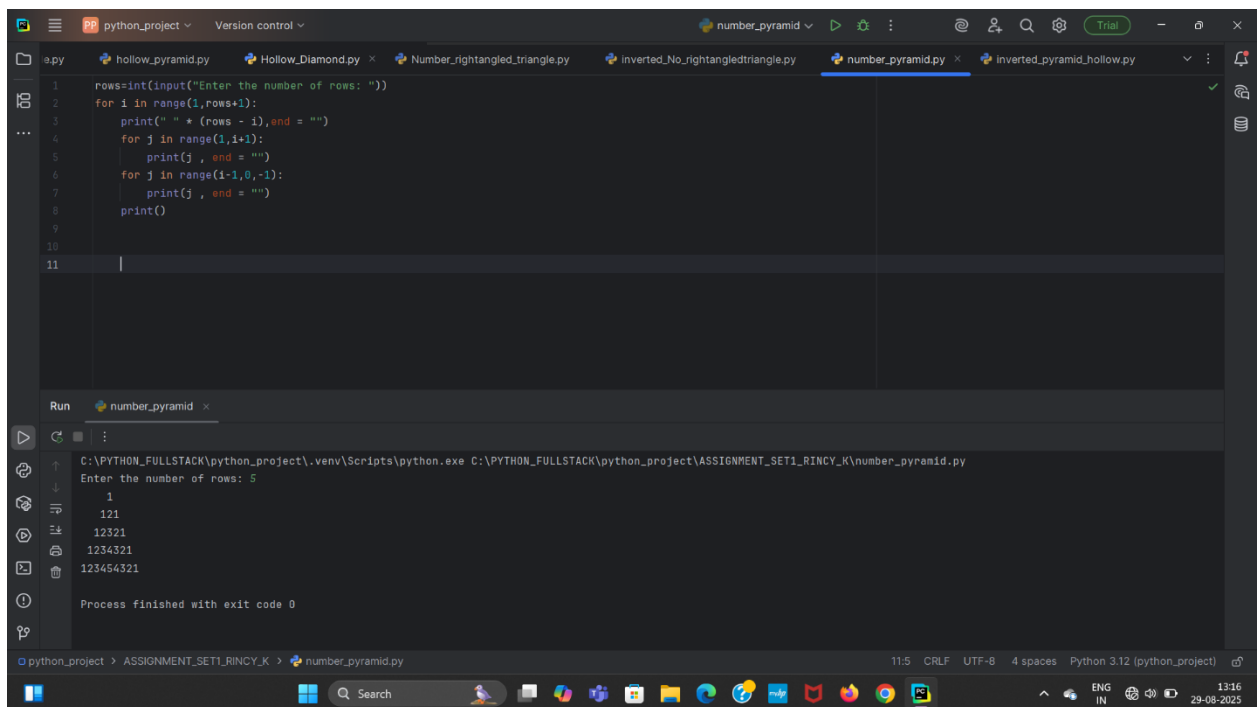
```
1 rows=int(input("Enter the number of rows :"))
2 for i in range(rows,0,-1):
3     for j in range(1,i+1):
4         print(j,end=" ")
5     print()
6
7
8
9
10
11
12
```

The Run console shows the following output:

```
Enter the number of rows :5
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Process finished with exit code 0

14. Number Pyramid



The screenshot shows a Python IDE with a file named `number_pyramid.py`. The code prompts the user to enter the number of rows, and when 5 is entered, it prints a number pyramid.

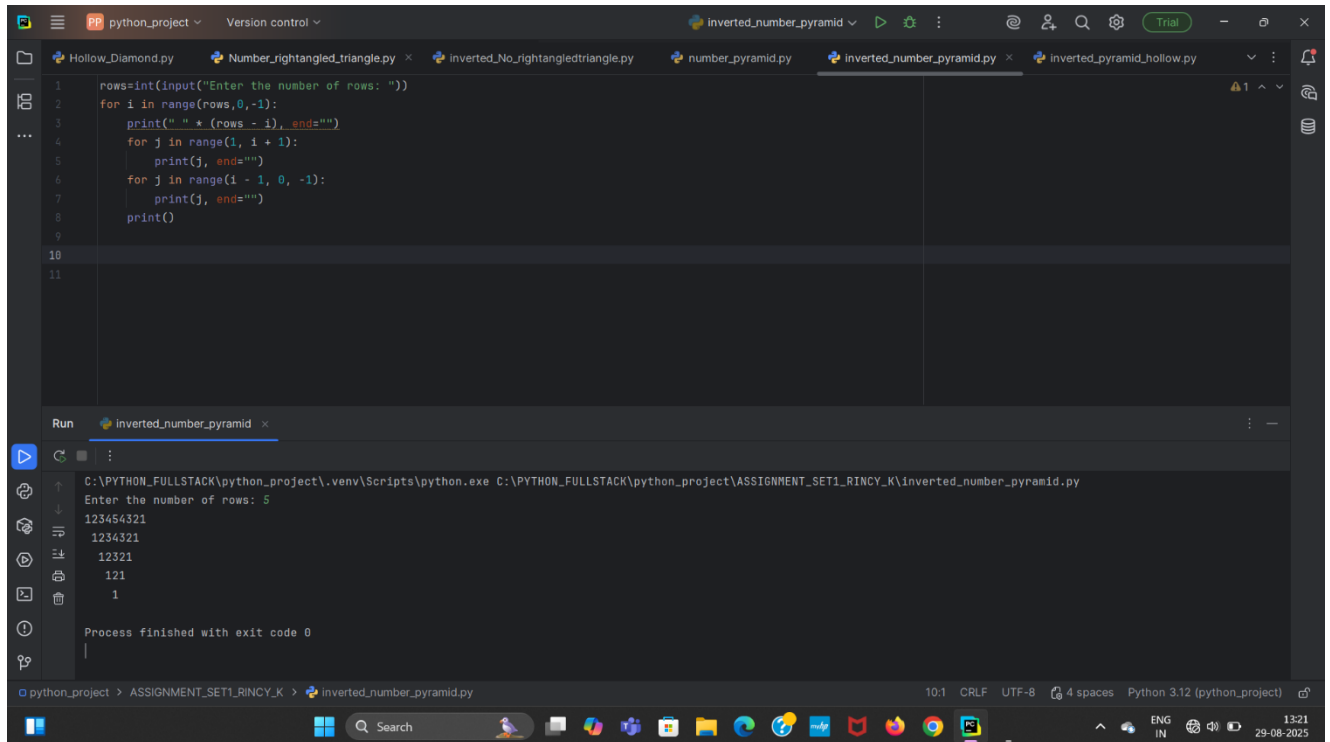
```
1 rows=int(input("Enter the number of rows: "))
2 for i in range(1,rows+1):
3     print(" " * (rows - i),end = "")
4     for j in range(1,i+1):
5         print(j , end = "")
6     for j in range(i-1,0,-1):
7         print(j , end = "")
8     print()
9
10
11
```

The Run console shows the following output:

```
Enter the number of rows: 5
1
121
12321
1234321
123454321
```

Process finished with exit code 0

15. Inverted Number Pyramid



The screenshot shows a Python IDE with a file explorer on the left, a code editor in the center, and a run console at the bottom. The code editor displays the following Python code for an inverted number pyramid:

```
1 rows=int(input("Enter the number of rows: "))
2 for i in range(rows,0,-1):
3     print(" " * (rows - i), end="")
4     for j in range(1, i + 1):
5         print(j, end="")
6     for j in range(i - 1, 0, -1):
7         print(j, end="")
8     print()
9
10
11
```

The run console shows the execution of the program with the input "5" and the resulting output:

```
C:\PYTHON_FULLSTACK\python_project\venv\Scripts\python.exe C:\PYTHON_FULLSTACK\python_project\ASSIGNMENT_SET1_RINCY_K\inverted_number_pyramid.py
Enter the number of rows: 5
123454321
 1234321
   12321
    121
     1
Process finished with exit code 0
```

The bottom status bar indicates the file path, encoding (UTF-8), and Python version (3.12).